P425/1
PRINCIPAL
MATHEMATICS
Paper 1
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3 Hours

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AITEL JOINT MOCKS ASSESSMENTS 2026

Uganda Advanced Certificate of Education

PRINCIPAL MATHEMATICS

PAPER 1

3 Hours

INSTRUCTIONS TO CANDIDATES

This examination paper consists of **two** *sections;* **A** *and* **B**. *It has* **five** *examination items.*

Section **A** *consists of* **one compulsory** *item.*

Section **B** *has two parts,* **I** *and* **II** *each with two optional items. Respond to only* **two** *items from Section B, choosing* **one** *item from each of the parts,* **I** *and* **II**.

Respond to **three** *items* **in all**.

Any additional item(s) responded to will **not** *be scored.*

Begin each item on a fresh page.

All *necessary working* **must** *be shown clearly.*

Graph paper is provided.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

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SECTION A

The item in this section is **compulsory**.

Item 1

An agricultural development board near the Guru industrial zone is planning to establish a secure perimeter and central infrastructure network to connect four regional storage and processing depots. On the district's Cartesian planning grid, the locations of the four depots are mapped as coordinates A (2, 6), B (5, 10), C (8, 6), and D (5, −2) respectively.

The project plan requires constructing straight direct boundary roads between adjacent depots to enclose the zone. Additionally, two main internal service pipelines must be laid diagonally across the zone to connect opposite depots (A to C, and B to D). To maximize efficiency, a central security and monitoring tower is to be built at the exact point where these two diagonal pathways intersect.

The structural engineering firm handling the project must mathematically verify the exact geometric shape of the enclosed area to qualify for specific regional infrastructure funding and determine the optimal coordinates for the monitoring tower.

Task:

a) Verify whether the boundary roads connecting the four depots enclose a region that forms a regular shape, and determine the precise coordinate location where the central monitoring tower should be built.

b) Calculate the total length of the internal service pipelines required to connect the opposite depots, and determine the equations of the lines representing these two diagonal pathways.

SECTION B

Answer only **two** *items from this section, choosing* **one** *item from each of the parts* **I** *and* **II**.

PART I

Item 2

A large-scale grain milling and processing factory near the Mbale industrial zone is conducting a comprehensive operational audit to maximize its financial efficiency and reduce overhead expenses. The plant manager needs to evaluate three critical sectors

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f the business: raw grain procurement optimization, automated hydration control ·ystems, and wholesale product inventory market readiness.

The factory's recent supply ledger shows three separate bulk batch purchases of Maize Flour, Millet Flour, and Sorghum Grain (measured in units of 100 kg bags) as detailed below:

Purchase Batch	Maize Flour (bags)	Millet Flour (bags)	Sorghum Grain (bags)
Batch A	2	1	2
Batch B	1	3	1
Batch C	3	1	1

The total financial expenditure for batches A, B, and C was 750,000/=, 1,000,000/=, and 800,000/= respectively. The management will classify future raw material acquisitions as cost-friendly if a prospective order of 5 bags of Maize Flour, 4 bags of Millet Flour, and 2 bags of Sorghum Grain stays below a total budget of 2,100,000/=.

To maintain precise product moisture levels, the factory uses an automated processing bay where a liquid conditioning compound is introduced. Sensor telemetry records indicate that the volume of conditioning fluid consumed during the daytime production shift is represented by $\sqrt{11x + 75}$ thousand litres, while the volume consumed during the night-time shift is $\sqrt{x + 25}$ thousand litres. The automated logs show that the total volume of fluid consumed over a full 24-hour cycle is exactly 20 thousand litres. The automated hydration system successfully protects the grains from spoilage without risking structural degradation only if the chemical variable x does not exceed 15 units.

Furthermore, the facility currently holds an inventory of 25 tons of processed flour sitting in its main storage reservoir. Due to regional market price volatility, the facility's inventory software implements a strict regulatory rule to automatically determine whether a stock batch is cleared for immediate external distribution or must be held back. A batch is cleared for export only if 90 times the number of tons in storage is strictly greater than the square of the tonnage plus 1400. The plant manager needs to verify if the current stock level satisfies this safety margin to be cleared for the market.

Task:

a) Formulate and solve the mathematical models representing the raw material procurement costs and the automated hydration system variables.

b) Evaluate the factory's inventory status against the regulatory rule and provide definitive recommendations to help the plant manager optimize operational decisions.

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Item 3

A commercial fruit processing factory near the Nateto industrial zone specializes in manufacturing concentrated juices. The factory relies on automated extraction chambers that operate under controlled conditions. The technical team observed that as the internal chamber pressure factor (x) increases, the efficiency percentage (E) of the extraction juice output decreases according to the polynomial model:

$$E=ax^3+bx^2+cx+40$$

Where a,b, and c are constant operation parameters determined by the fruit density. The quality control log shows the following efficiency levels recorded at different pressure factors using the same batch configuration:

Pressure Factor (x)	Extraction Efficiency (E)
1	37
2	32
3	19

An external production consultant recommends that the factory should aim to run the extraction process at a pressure factor of $x = 4$ to conserve power, claiming it will still maintain a viable efficiency.

To streamline the workflow before the pasteurization phase, the factory employs two heavy-duty extraction pumps, Pump P and Pump Q. The operational workloads of these pumps depend on the running time, t minutes, according to the mathematical models:

Pump P: 32^{t-1}

Pump Q: $28(3^t) - 9$

To prevent fluid congestion in the pipeline, both pumps must achieve a perfect equilibrium of workloads, and the operational time t required to reach this balance should not exceed 3 minutes.

Additionally, the factory stores its finished product in a multi-level automated warehouse consisting of 9 distinct storage zones. The total capacity of these zones is represented by the terms of the binomial expansion:

$$N=(x+3)^8 \text{ hundred crates}$$

Where x represents the baseline clearance height (in meters) allocated for stackable shipping crates. The factory management plans to expand their current distribution orders. This expansion will be approved only if the storage capacity calculated using

the first four terms of the expansion with crates set at a clearance height of x=0.2 m shows that the facility can accommodate at least 550,000 crates.

Task:

a) Help the factory officials to get the system of equations from the efficiency log to determine the constants a, b, and c, and analyse whether the consultant's recommended operating pressure yields a viable efficiency.

b) Determine the operational time required for the two pumps to achieve equilibrium, and evaluate whether the warehouse capacity meets the criteria needed to approve the factory's distribution expansion plans.

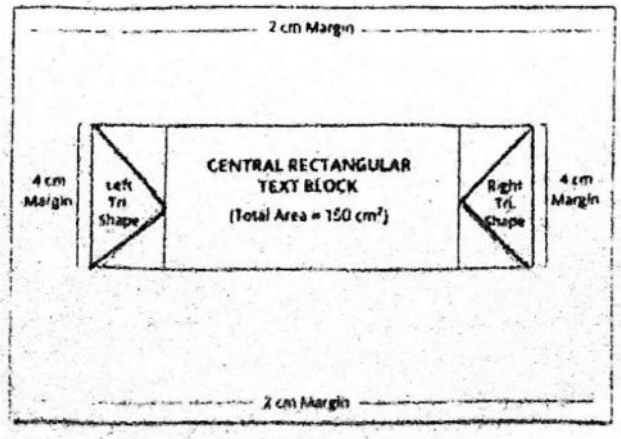
PART I

Item 4

An agro-environmental cooperative based in Mbarara City designs specialized concrete water basins to harvest clean rainwater for livestock during drought seasons. The internal chamber of a newly proposed basin model is engineered by rotating a two-dimensional region bounded by the horizontal line $y = 6$ and the curve $y = x^2 + 2$ about the line $y = 6$.

To secure a localized green energy grant, the cooperative must print a standard technical brief document. The layout specifications dictate that the central text area dedicated to the engineering metrics must be shaped as a composite block: it consists of a central rectangle flanked by two identical right-angled triangles on its left and right sides to highlight the structural cross-sections. This entire combined central text block must cover an exact area of 150 cm^2.

To guarantee clear binding and readable presentation margins, the document layout requires a fixed margin of 4 cm on both the left and right sides of the paper, and a fixed margin of 2 cm at both the top and bottom.



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Mbarara District print-office guidelines state that for the document to fit standard evaluation folders, its total combined horizontal width (including both the inner text blocks and the side margins) must be kept within a strict limit between 18 cm and 35 cm. The cooperative wants to optimize the layout by finding the precise external dimensions that minimize the total combined surface area of the paper sheet to keep paper costs low, while preserving the mandatory central text block space.

Additionally, before submission, the team must compute and explicitly print the exact volume capacity of the water basin on the document sheet.

Task:

a) Determine the optimal external dimensions of the paper sheet that minimize its total area while meeting the print-office layout limits.

b) Calculate the exact volume capacity of the water basin to complete the technical brief.

Item 5

An organic juice refinery situated near Soroti City is upgrading its packaging line to handle increased production volumes. The plant manager plans to channel the refined fruit concentrate into specialized cylindrical canisters designed such that the height of each canister is exactly four times its radius. The canister fabrication plant has a known manufacturing tolerance error in the radius of 0.5%. The refinery manager will reject the batch of canisters if the corresponding percentage error in the total volume exceeds 1.6%.

The automated induction nozzle intended for the filling process records the instantaneous height (h) and the volume (V) of the fluid inside the canister after an elapsed time of t seconds according to the following models:

$$\mathrm{h} = \left(\frac{t^2}{4} + 2t\right) \text{ cm and } \left(\frac{t^2+6t}{t+2}\right) \text{ cm}^3$$

The technical team will approve the operation of this nozzle only if the rate of change of the fluid volume with respect to its height at the exact time t=2 seconds does not exceed 0.5 cm^2. This threshold is strictly enforced to prevent fluid turbulence and splashing.

Furthermore, the weekly operational energy cost of running the filling system is modelled by the differential equation:

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$$\frac{dc}{dt} = 30te^{-\frac{t}{4}}$$

Where c represents the cumulative maintenance and energy cost in thousands of Ugandan shillings, and t represents the continuous processing duration in weeks. The finance department will transition the facility to an alternative solar hybrid grid if the total accumulated cost of running the system over the first 4 weeks exceeds 180,000/=.

Task:

a) Analyze the geometric tolerance dimensions of the canisters and determine whether the filling nozzle configuration satisfies the non-spill criteria at the specified time interval.

b) Compute and solve the cost differential equation to advise the finance department on whether the refinery needs to switch to the alternative solar hybrid grid based on the four-week budget limit.

END

$(x-9)^2 + (y-5)^2 = r^2$

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